

Global high-order polarization tails for bounded interval polynomials

Auxiliary proof; independent mathematical reviews completed

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The full argument passed separate mathematical reviews by the rational-design and independent-directions collaborators; this is not formal verification. This note supplies the global polynomial tail estimate used in the finite-Jackson endpoint-profile argument. It asserts only the polynomial derivative and high-order tail estimates below, not a new profile theorem.

Lemma 1 (All-order interior Bernstein bound). *Let P be a polynomial of degree at most $d \geq 1$, with $\|P\|_{L^\infty[0,T]} \leq U$. Then, for $0 < v < T$ and $1 \leq k \leq d$,*

$$\frac{|P^{(k)}(v)|}{k!} \leq U \left(\frac{Cd}{k\sqrt{v(T-v)}} \right)^k, \quad (1)$$

where C is absolute.

Proof. First work on $[-1, 1]$. For z outside the interval put $g(z) = \log |z + \sqrt{z^2 - 1}|$, choosing the root so the modulus is at least one, and set $g = 0$ on the interval. If $x \in (-1, 1)$, $a = \sqrt{1 - x^2}$, and $|z - x| \leq R$, then

$$g(z) \leq C\sqrt{R}, \quad g(z) \leq CR/a. \quad (2)$$

For the first bound when $R \leq 1$, compare a square root of $z^2 - 1$ to the closer of the two square roots of $x^2 - 1$. Their distance is at most $\sqrt{|z^2 - x^2|} \leq C\sqrt{R}$, since the product of the two candidate distances is $|z^2 - x^2|$. Both boundary values $x \pm \sqrt{x^2 - 1}$ have modulus one. Thus $e^{g(z)} \leq 1 + R + C\sqrt{R}$, which proves the bound. For $R \geq 1$, $g(z) \leq \log(C(1 + R)) \leq C\sqrt{R}$. If $R \leq a^2/8$, a local analytic branch of $\log(z + \sqrt{z^2 - 1})$ has derivative of modulus $|z^2 - 1|^{-1/2} \leq C/a$ on the disk, because its distance from each endpoint is comparable to the corresponding distance of x . Integration from x proves the second bound in this case. Otherwise the first bound and $R > a^2/8$ prove it.

For completeness, the Bernstein–Walsh estimate here follows directly from the maximum principle: if Q is bounded by U on $[-1, 1]$, then $w^{-d}Q((w + w^{-1})/2)$ is analytic for $|w| > 1$, including infinity, and bounded by U on $|w| = 1$. Hence $|Q(z)| \leq Ue^{dg(z)}$. Cauchy’s formula on the circle of radius $R = ka/d$, using (2), gives $|Q^{(k)}(x)|/k! \leq U(Cd/(ka))^k$. Rescaling $x = 2v/T - 1$ proves (1). $\square$

Use the marked-tensor notation

$$M = m - 1, \quad \ell = \lfloor M/4 \rfloor, \quad N = M - \ell, \quad z_j = \ell(1 - p_j), \quad s = \sum_{j \neq i} (1 - p_j).$$

For a uniform ℓ -subset B of the unmarked indices, let $A_B = \prod_{j \in B} p_j$, let R be its complement, and put

$$v = N^{-1} \sum_{j \in R} z_j, \quad a_k = e_k((z_j - v)_{j \in R}) / \binom{N}{k}.$$

We use the established endpoint balance and Laplace row bound

$$\max_j z_j \leq C(s+1), \quad \mathbb{E}_w e^{-cs} \leq C_c/m \quad (c > 0 \text{ fixed}). \quad (3)$$

The marked column is included in the maximum. The elementary centered-coefficient estimate gives

$$|a_k| \leq \left(\frac{ek\sigma^2}{N^2} \right)^{k/2}, \quad \sigma^2 = \sum_{j \in R} (z_j - v)^2. \quad (4)$$

Theorem 1 (Negligible high differential orders). *Fix $A, C_0 > 0$. There are constants $\kappa > 0$ and $B > 0$ such that the following holds for all sufficiently large m . Let $d = \lfloor \kappa m \rfloor$, $J = \lceil B \log m \rceil$, and let P have degree at most d with $\|P\|_{L^\infty[0, \ell]} \leq m^{C_0}$. Then*

$$(\ell + 1) \mathbb{E}_w \mathbb{E}_{BA_B} \sum_{k=J+1}^{d+1} |a_k| \left[(1 + |z_i - v|) \frac{|P^{(k)}(v)|}{k!} + \frac{|P^{(k-1)}(v)|}{(k-1)!} \right] \leq m^{-A}.$$

Derivatives beyond the degree vanish. This estimate covers both the unmarked and direct marked polarization remainders.

Proof. Write $U = \|P\|_\infty$. First restrict to subsets with $v \leq \ell/2$. From (3), $\sigma^2 \leq \sum_{j \in R} z_j^2 \leq CN(s+1)v$. Combining (1) and (4), for $k \geq 2$, gives

$$|a_k| \frac{|P^{(k)}(v)|}{k!} \leq U \left(\frac{Cd}{m} \sqrt{\frac{s+1}{k}} \right)^k. \quad (5)$$

Here $\ell - v \geq \ell/2 \asymp m$ and $N \asymp m$. The formulas extend to $v = 0$: then all complementary z_j vanish, so $a_k = 0$ for $k \geq 1$.

For each row, Maclaurin's inequality gives $\mathbb{E}_B A_B \leq e^{-\ell s/M}$. From (3) it follows that for every $h \geq 0$ and fixed $c > 0$,

$$(\ell + 1) \mathbb{E}_w e^{-cs} (s+1)^h \leq C^{h+1} \Gamma(h+1). \quad (6)$$

Use (6) on (5), including the extra factor $1 + |z_i - v| \leq C(s+1)$. Stirling's bounds show that the resulting integrated k th term is at most $CU(k+1)^3 (Cd/m)^k$. For the derivative of order $k-1$, the same calculation leaves one factor $\sqrt{k(s+1)v/N}$. Since $v \leq Cs$, its integrated bound is

$$CU(k+1)^3 m^{-1/2} (Cd/m)^{k-1}. \quad (7)$$

Constants can change from line to line. These estimates also cover $k = d+1$, whose derivative of order k is zero.

On the remaining subsets $v > \ell/2$, use instead the universal variance bound

$$\sigma^2 \leq Nv(\ell - v),$$

which follows from $0 \leq z_j \leq \ell$ and $\sum z_j^2 \leq \ell \sum z_j$. Equations (1) and (4) therefore give

$$|a_k| \frac{|P^{(k)}(v)|}{k!} \leq U \left(\frac{Cd}{\sqrt{Nk}} \right)^k \leq U \exp(Cd^2/N). \quad (8)$$

The last inequality follows by maximizing $(k/2) \log(Cd^2/(Nk))$ over positive real k . For the derivative of order $k-1$, the same argument costs at most an additional polynomial factor in m , since $\sqrt{v(\ell - v)} \leq \ell/2$. At $v = \ell$ all a_k for $k \geq 1$ vanish, so no endpoint exception is needed.

The condition $v > \ell/2$ forces $s > N/2$, because $v \leq \ell s/N$. Hence the success-weighted contribution of these rows is at most a polynomial factor times

$$U \exp(Cd^2/N - cm).$$

Indeed one can first bound each summand by (8), then average A_B using $\mathbb{E}_B A_B \leq e^{-\ell s/M}$ and finally use $\mathbb{E}_w 1 = 1$. Choose the fixed κ small enough to make this exponentially small in m .

Decrease κ further so the ratios in (5)–(7) after integration are at most a fixed $\rho < 1$. The sum of those bounds over $k > J$ is at most $CU \sum_{k>J} (k+1)^3 \rho^{k-1}$. Since $U \leq m^{C_0}$, choosing B sufficiently large in terms of A, C_0 makes the sum at most half the required bound. The exponentially small part is eventually smaller than the other half. $\square$

This argument is entirely global over the rows and does not require an interior lower cutoff on their deficit sum. To remove the logarithm in the profile rate, one still needs a positive polynomial kernel of width $m^{-1/2}$ with localized derivatives through order J .